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Changing SQL Server Collation After Installation

By: Douglas P. Castilho | Read Comments (39) | Related Tips: 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | More > SQL Server Configurations

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Problem

In some scenarios we can find different SQL Server collations between the server instance and its databases. Sometimes the collation is fixed in some table columns or inside stored procedures to solve some of relationship problems with columns that have different collations. The most common technique to have a consistent collation for the instance and all it databases has a lot of steps and can sometimes be problematic. Can you provide any suggestions?

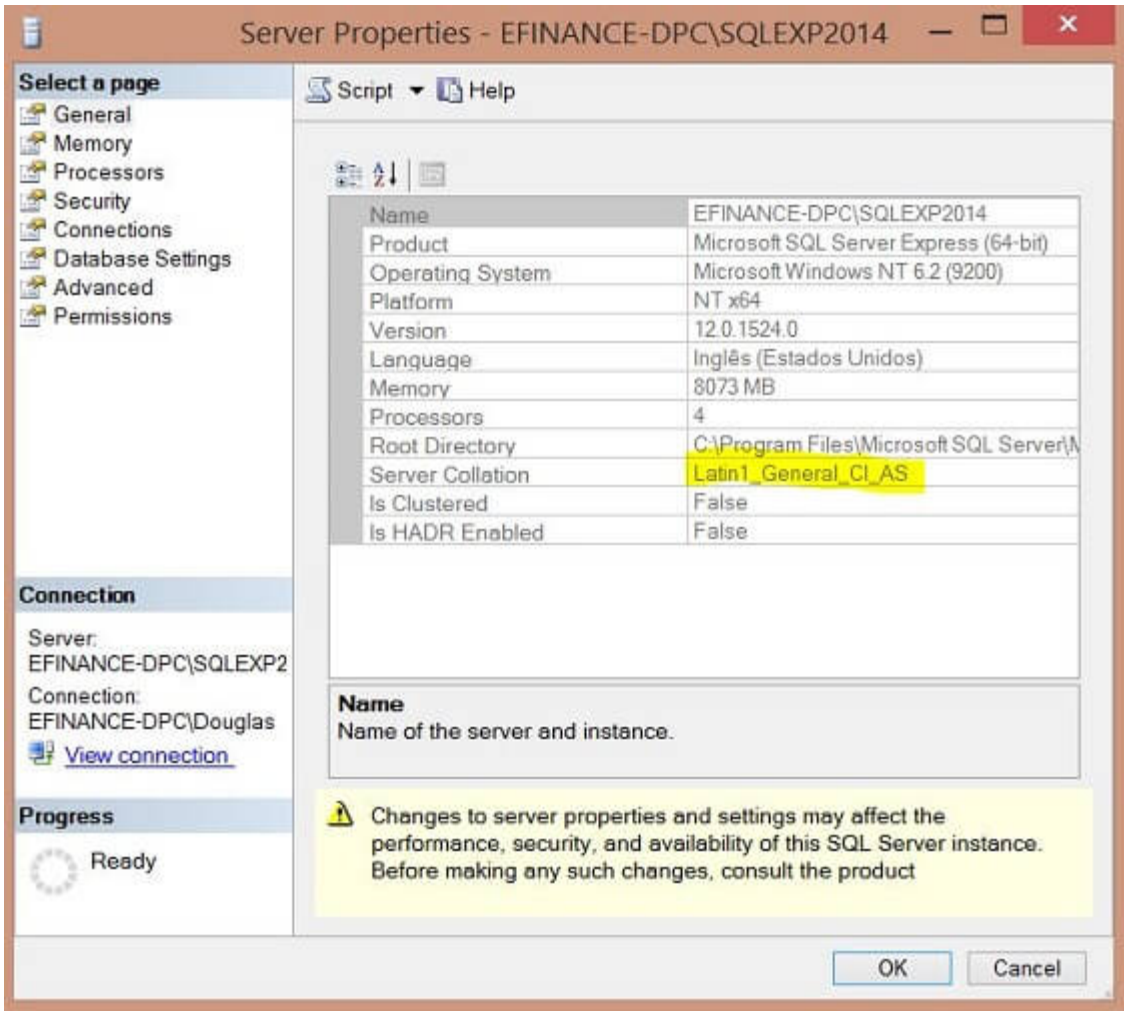
Solution

A simple way to correct the collations in a few steps is outlined in this tip. It is recommended **to create a backup of all databases (including system databases)** before take administrative actions on a SQL Server instance. It is important to ensure that there is no fixed collation logic in columns or inside stored procedures, triggers, etc., otherwise the command below may report problems.

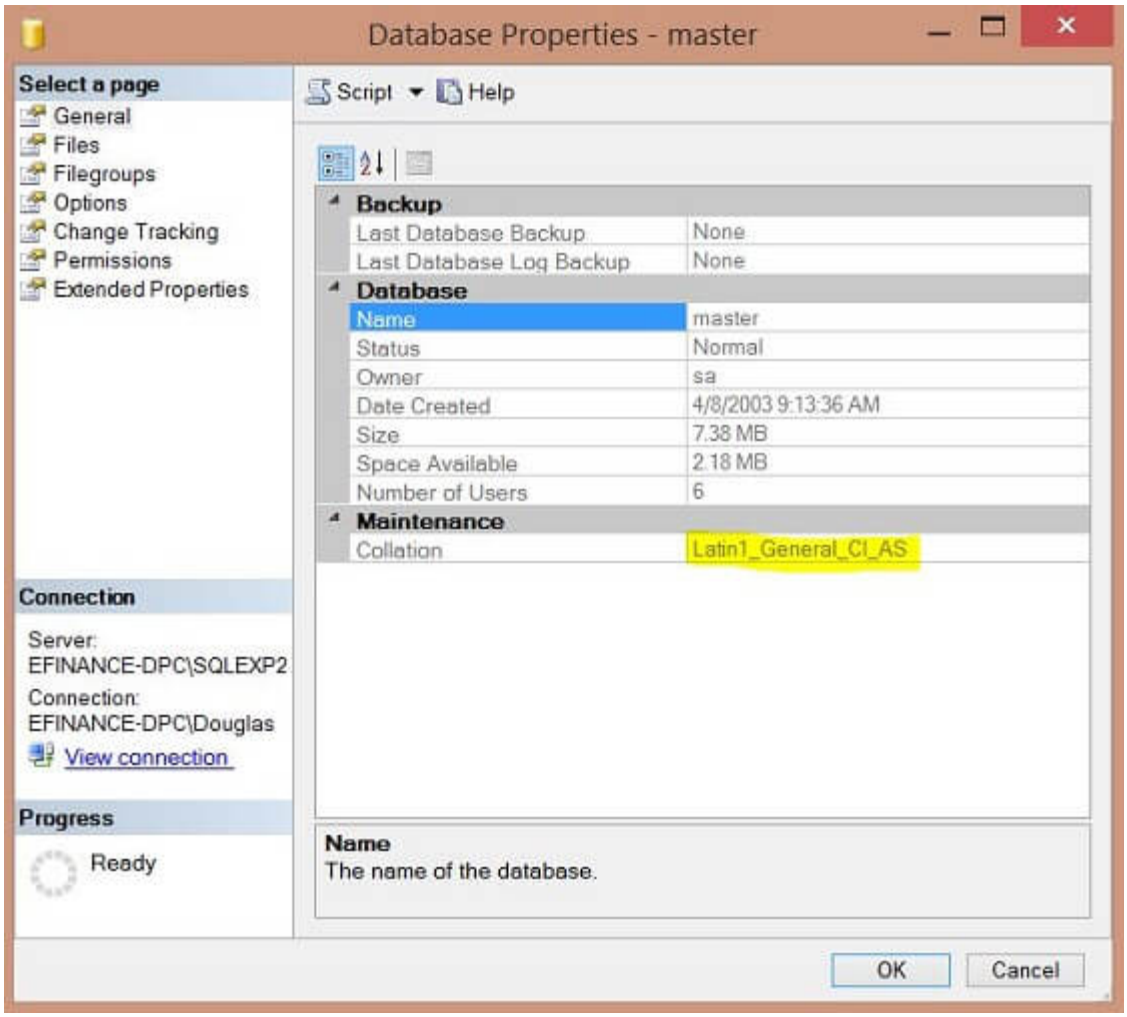
Step 1 - Determine the SQL Server Collation

Let's confirm the current SQL Server instance collation and all it databases including system databases collation before taking actions.

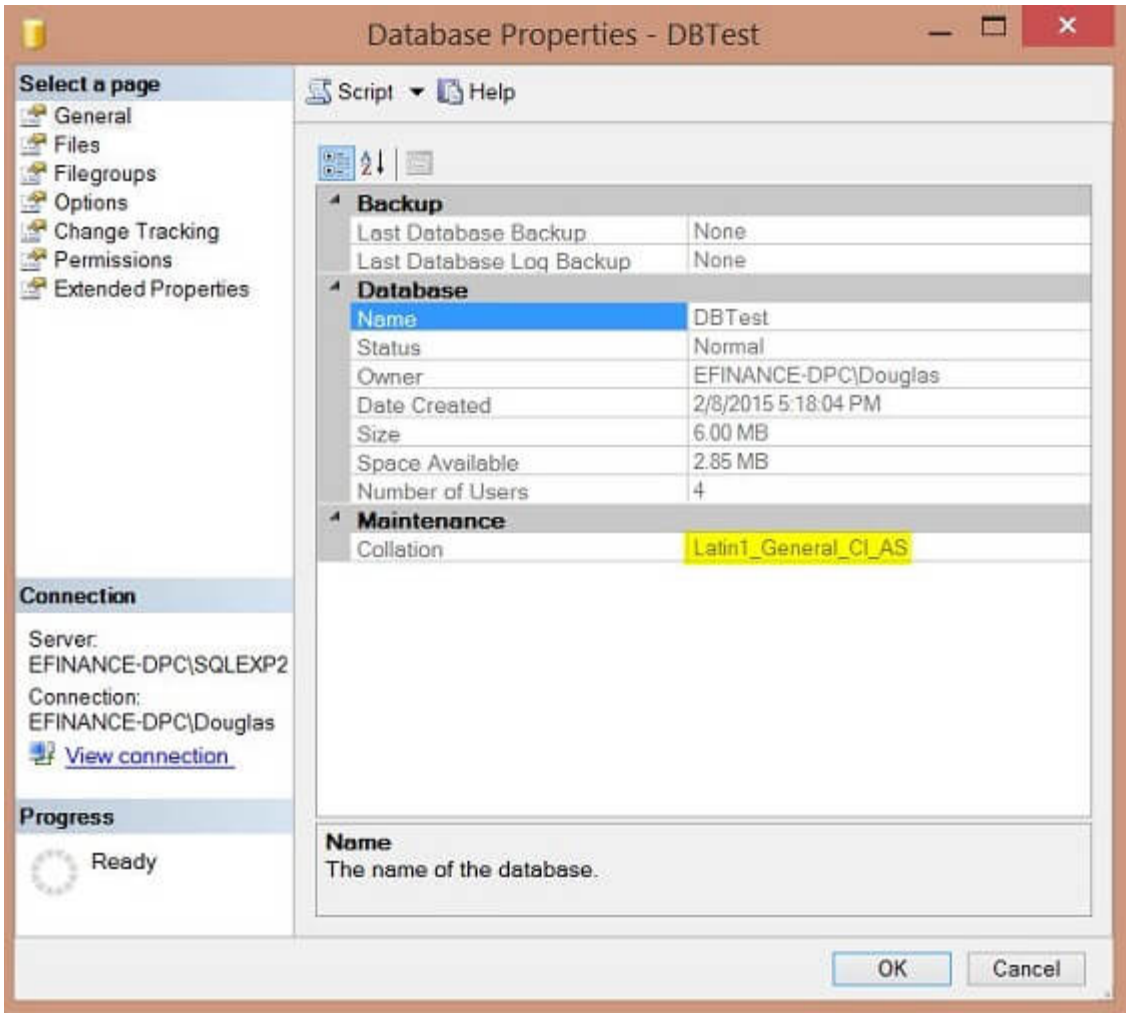
SQL Server Instance Collation



SQL Server Master Database Collation



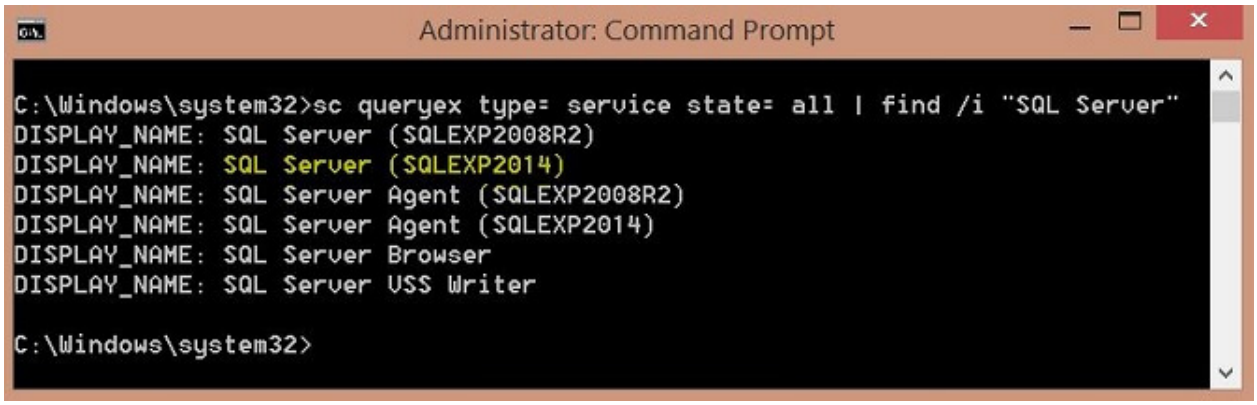
SQL Server DBTest Database Collation



The server has the "Latin1_General_CI_AS" collation and we'll change it to "SQL_Latin1_General_CP1_CI_AI" for this test.

Step 2 - Stop the SQL Server Services

First of all, we have to stop the SQL Server Services for the changes to be applied.



```
Administrator: Command Prompt
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Windows\system32>NET STOP "SQL Server (SQLEXP2014)"
The SQL Server (SQLEXP2014) service is stopping.
The SQL Server (SQLEXP2014) service was stopped successfully.


C:\Windows\system32>
```

Step 3 - Open a Command Prompt and Navigate to the Binn Directory

Now we have to open a command prompt with administrative privileges and go to the BINN directory of Microsoft SQL Server, following the example below:

```
Administrator: Command Prompt

C:\Program Files\Microsoft SQL Server\MSSQL12.SQLEXP2014\MSSQL\Binn>dir sqx.exe
 Volume in drive C has no label.
 Volume Serial Number is F2A3-DC39

 Directory of C:\Program Files\Microsoft SQL Server\MSSQL12.SQLEXP2014\MSSQL\Bin
n

10/04/2013  12:17 PM           612,544 SQLAGENT.EXE
10/04/2013  12:17 PM         2,806,464 SQLIOSIM.EXE
10/04/2013  12:17 PM           104,640 sqlmaint.exe
10/04/2013  12:17 PM           370,368 sqlservr.exe
10/04/2013  12:17 PM            26,304 sqlstubss.exe
               5 File(s)      3,920,320 bytes
               0 Dir(s)  23,478,476,800 bytes free

C:\Program Files\Microsoft SQL Server\MSSQL12.SQLEXP2014\MSSQL\Binn>
```

This picture shows the SQL Server Binn directory and "sqlservr.exe" that will be used in this test.

Step 4 - Apply a New SQL Server Collation

Execute the command below. A lot of information will appears and no user action is required, just close the prompt window after the execution ends. The parameter "-s" is only necessary if more than one SQL Server instance exists on the target machine.

```
sqlservr -m -T4022 -T3659 -s"SQLEXP2014" -q"SQL_Latin1_General_CP1_CI_AI"
```

Parameters used:
[-m] single user admin mode
[-T] trace flag turned on at startup
[-s] sql server instance name
[-q] new collation to be applied

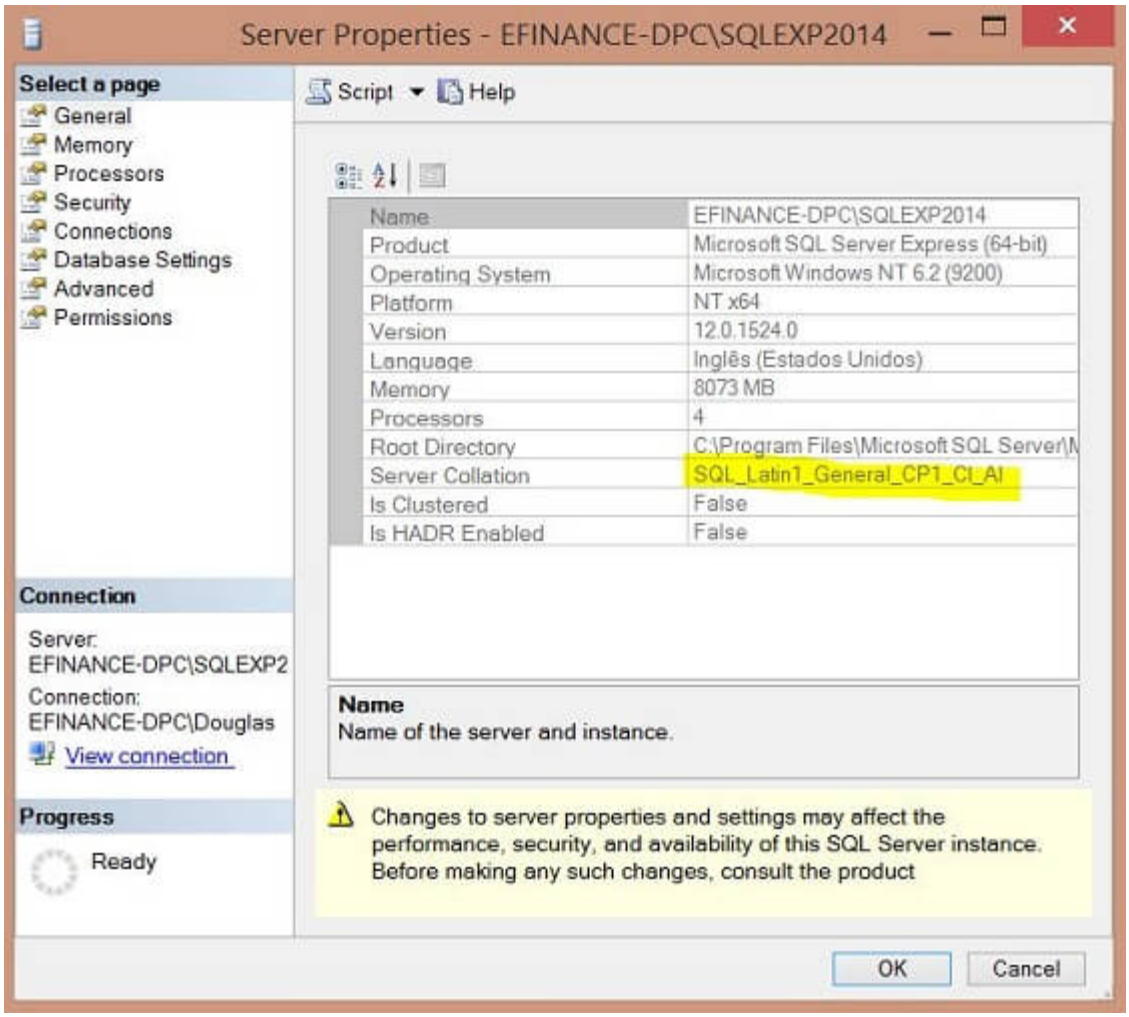
```
Administrator: Command Prompt - sqlservr  -m -T4022 -T3659 -s"SQLEXP2014...
2015-02-08 17:34:08.82 spid7s       index restored for DBTest.sysxprops.
2015-02-08 17:34:08.82 spid7s       index restored for DBTest.sysscalartypes.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysidxstats.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysendpts.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysclsobjs.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysremsvcbinds.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysrts.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.sysasymkeys.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.syssqlguides.
2015-02-08 17:34:08.83 spid7s       index restored for DBTest.syssoftobjrefs.
2015-02-08 17:34:08.83 spid7s      The default collation was successfully change
d.
2015-02-08 17:34:08.84 spid7s      Recovery is complete. This is an informationa
l message only. No user action is required.
```

Step 5 - Start SQL Server and Verify Collation

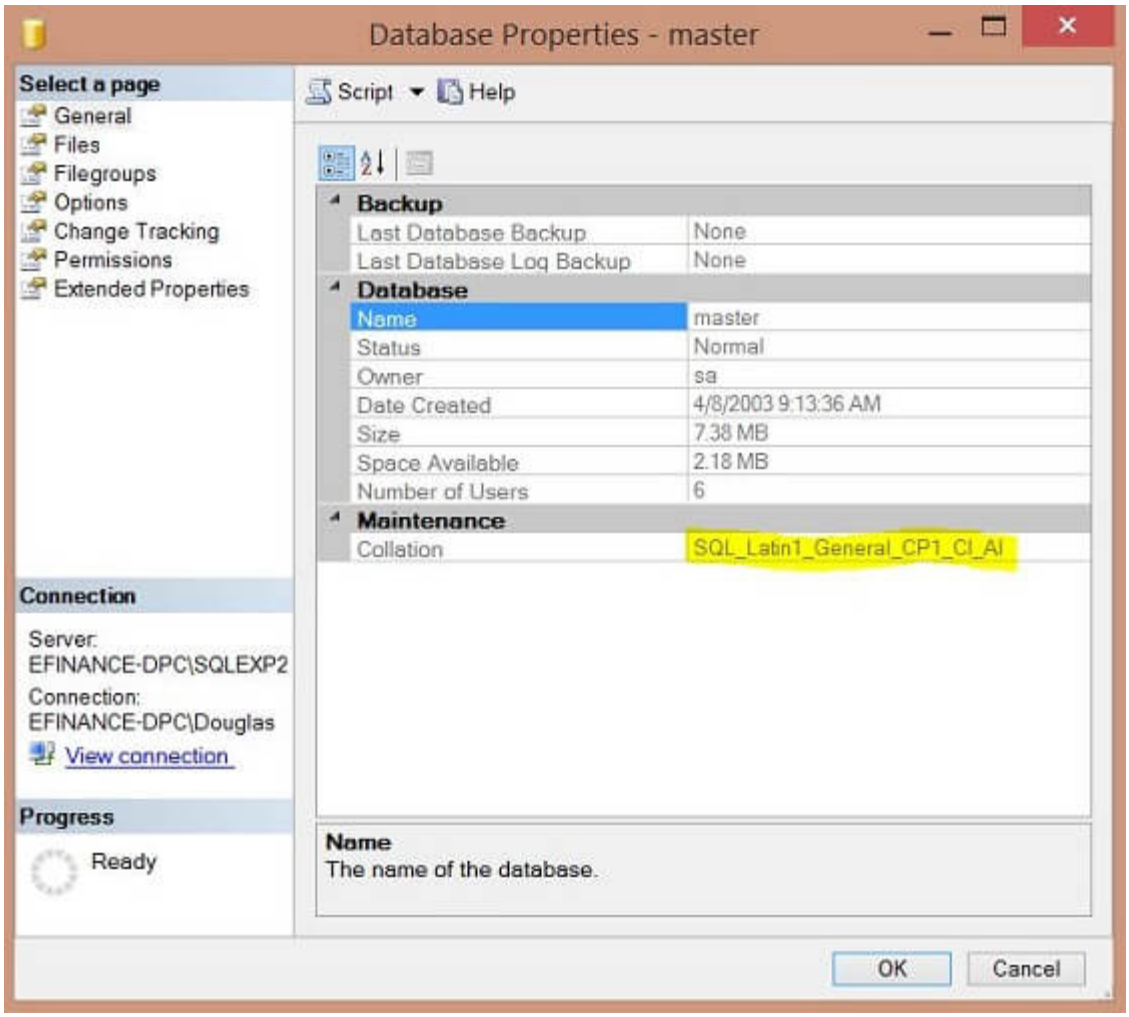
Start the SQL Server instance and confirm that all changes were applied on the instance and all databases including system databases have the correct collation.

```
Administrator: Command Prompt
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Windows\system32>NET START "SQL Server (SQLEXP2014)"
The SQL Server (SQLEXP2014) service is starting.
The SQL Server (SQLEXP2014) service was started successfully.


C:\Windows\system32>
```

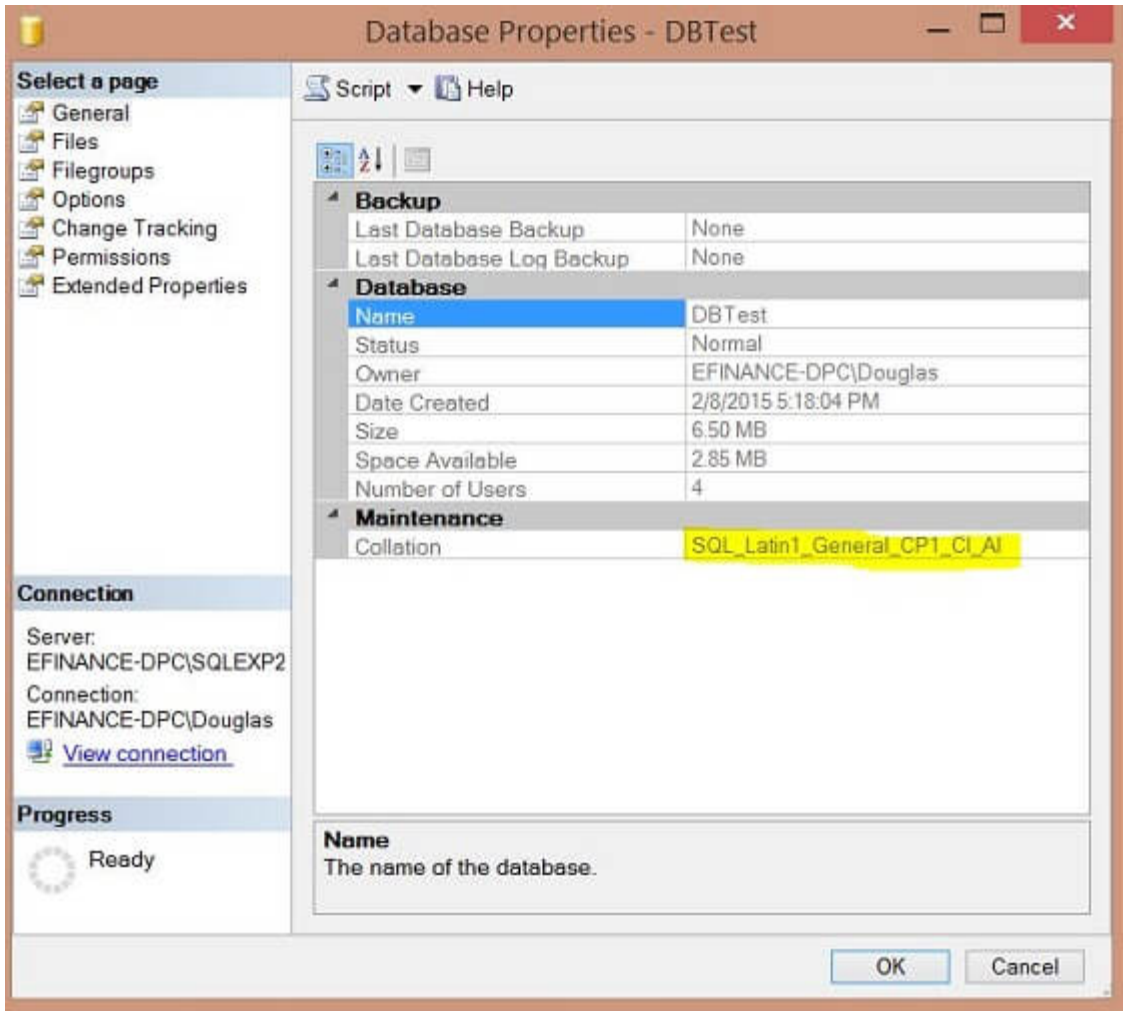
SQL Server Instance New Collation



SQL Server Master Database New Collation



SQL Server DBTest Database New Collation



Next Steps

- Does this tip sound interesting to you?Please check out these tips below to explore additional scenarios:
  - How to change server level collation for a SQL Server Instance
  - How to Change the COLLATION of a SQL Server Column

- Keep in mind this is an undocumented solution and may not be supported by Microsoft if you have issues, so please test carefully before implementing in a production environment.

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About the author

Douglas Castilho has been a SQL Server DBA over 6 years, focuses on tuning, backup, disaster recovery, mirroring, T-SQL, PL-SQL and .NET

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